

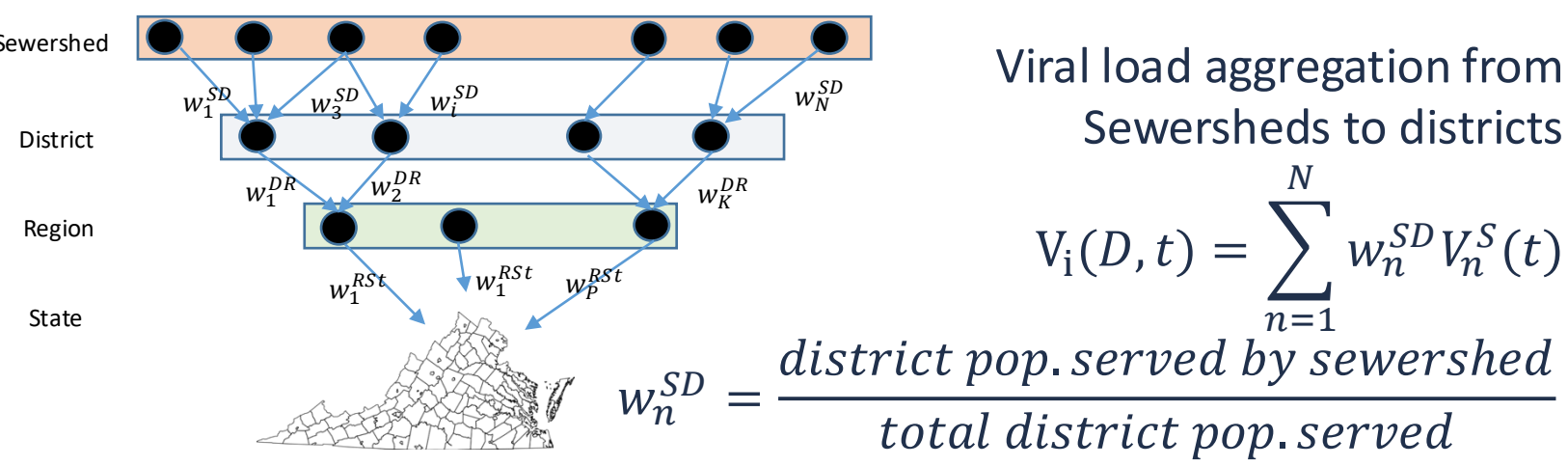
Interplay and Predictive Power of COVID-19 Incidence Indicators in Virginia

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Motivation

- To investigate the relationship between wastewater data and various COVID-19 related indicators such as Hospitalizations, Cases, COVID-Like Illness (CLI) by utilizing quantile (level-based) categorization.
- Aim: to develop a predictive model for cases and hospitalizations (categories), while evaluating forecast performance based on different signals used in the model, time frames, and evaluation metrics.

Dataset Details & Spatiotemporal Alignment



Categorical Indicators

Level-based categories: determining bin boundaries based on 3 quantiles.

- 1 = Low,
- 2 = Med,
- 3 = High.

Test of association

- Chi-square test:** to measure the significance of the association.
 H_0 : Cases are independent of Viral loads
- Test Statistics:
$$\chi^2 = \sum_{i,j} \frac{(O_{i,j} - E_{i,j})^2}{E_{i,j}}; E_{i,j} = \frac{r_i c_j}{N}$$

Based on the observed data we get
 $\chi^2_{df=4} = 566.03$, p-value < 0.05,
Cohen's $\omega = 0.52$
- Chi-square test shows that cases and wastewater have a strong association.

OBSERVED		Viral Load			
Cases		Low	Med	High	Total
Low	446	192	68	706	
Med	189	281	222	692	
High	71	219	398	688	
Total	706	692	688	2086	

EXPECTED		Viral Load			
Cases		Low	Med	High	Total
Low	239	234	233	706	
Med	234	230	228	692	
High	233	228	227	688	
Total	706	692	688	2086	

Multinomial regression formulation

We consider the multinomial regression model to forecast case and hospitalization categories. For instance, to fit a model

$$\eta(Q_{cases}(t)) = \beta_0 + \beta_1 Q_{VL}(t-1),$$

we train 2 logistic regression models for $k = 2, 3$

$$\eta(Q_{cases}(t)) = \log\left(\frac{P(Q_{cases}(t) = k)}{P(Q_{cases}(t) = 1)}\right) = \beta_{0,0}^k + \beta_{1,1}^k I_{Q_{VL}(t-1)=1} + \beta_{1,2}^k I_{Q_{VL}(t-1)=2} + \beta_{1,3}^k I_{Q_{VL}(t-1)=3}$$

here $Q_{cases}(t)$ denotes the quantile-category for cases at time t and $I_{Q_{cases}(t-1)=i}$ is the indicator variable for the event $\{Q_{cases}(t-1) = i\}$.

Forecast model: incorporating multiple signals

- Experiment Setup:
- Time period: Sept 2021 to June 2023, starting with 20 weeks for model training, for each district.
 - Real-time forecasting: updated quantile categories with each new data point.
 - Forecast horizon: 1-week ahead forecasts for all districts.

The precision of combined district-level category forecasts is **significantly improved** by incorporating hospitalizations and CLI into the model.

- ✓ 11% improvement in average precision
- ✓ 13% increase in accuracy within the “High” category

Model 1:

$$\eta_{Q_{cases}(t)} = \beta_0 + \beta_1 Q_{VL}(t-1)$$

Model 1		Predicted Categories			
		Low	Med	High	Total
Actual categories	Low	446	192	68	706
		63.2%	27.7%	9.9%	33.8%
	Med	189	281	222	692
		26.8%	40.6%	32.3%	33.2%
High	71	219	398	688	
		10%	31.7%	57.8%	33%
		706	692	688	2086
		100%	100%	100%	

Average precision = 53.93%

Model 2:

$$\eta_{Q_{cases}(t)} = \beta_0 + \beta_1 Q_{VL}(t-1) + \beta_2 Q_{hosp}(t-1) + \beta_3 Q_{CLI}(t-1)$$

Model 2		Predicted Categories			
		Low	Med	High	Total
Actual categories	Low	531	148	27	706
		75.2%	21.4%	3.9%	33.8%
	Med	156	335	201	692
		22.1%	48.4%	29.2%	33.2%
High	25	173	490	688	
		3.5%	25.0%	71.2%	33%
		712	656	718	2086
		100%	100%	100%	

Average precision = 65%

Results – Key Takeaways

- At the category level, viral load, hospitalizations, and CLI, each have a strong association with cases.
- Forecast performance depends on data or the time frame used for the model training.
- Adding other important signals such as CLI and hospitalizations improves model precision.
- Forecast model performance is similar for cases and hospitalizations.
- In most districts, probability forecasts outperform the Naïve model (uniform).

Evaluation Metrics

- To evaluate the probability forecasts we use **Brier score**:

$$BS(D, t) = \sum_{i=1}^3 (p_i(D, t) - o_i(D, t))^2$$

here $p_i(D, t) = P(Q_{cases}(D, t) = i)$ and $o_i(D, t)$ is the observed probability, for district D at time t .

- To evaluate the point forecasts $\widehat{Q}_{cases}(D, t) = \arg \max_i p_i(D, t)$, we use the **precision** metric.

$$\text{Precision}(i) = \frac{\text{Number of correctly classified forecasts in category } i}{\text{Total number of forecasts in category } i}; i \in \{Low, Med, High\}$$

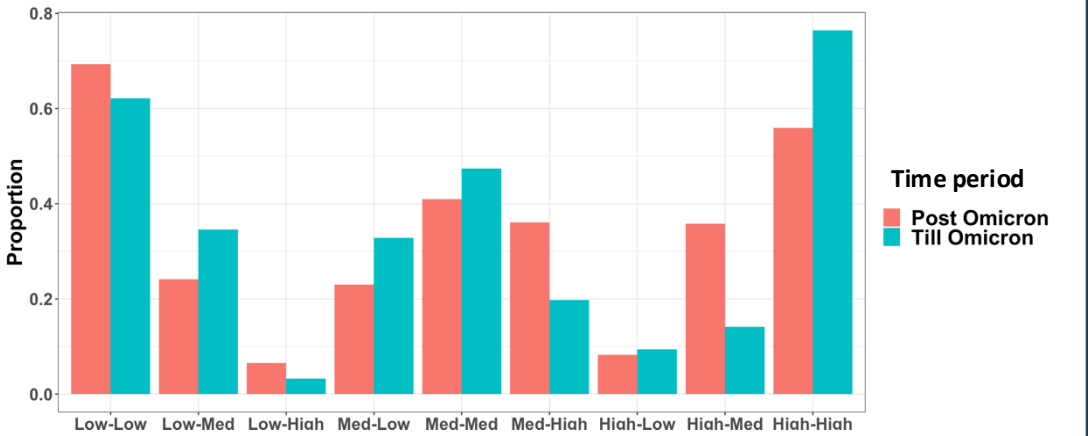
Impact of time-frame used for analysis and forecast models:

Before and after Omicron wave

- Precision of the forecast model (incorporating all signals) decreases after the Omicron wave.

	Precision Low	Precision Med	Precision High	Average precision
Till Omicron	89.7%	64.6%	83.2%	79.6%
Post Omicron	76.4%	54.2%	62%	64.3%

- The association between Cases and Wastewater remains statistically significant (based on χ^2 test) post Omicron.



Comparison between Case forecasts & Hospitalization forecasts

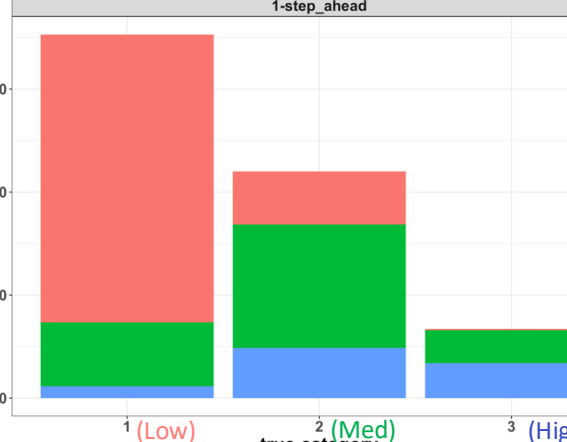
We use a multinomial model with all signals:

$$\eta(Q_{cases}(t)) = \beta_0 + \beta_1 Q_{cases}(t-1) + \beta_2 Q_{hosp}(t-1) + \beta_3 Q_{CLI}(t-1) + \beta_4 Q_{VL}(t-1).$$

A similar model is used to model $Q_{hosp}(t)$.

- Combining forecasts from all districts, the average precision is similar for hospitalizations and cases.

Forecasts for cases



Forecasts for hospitalizations

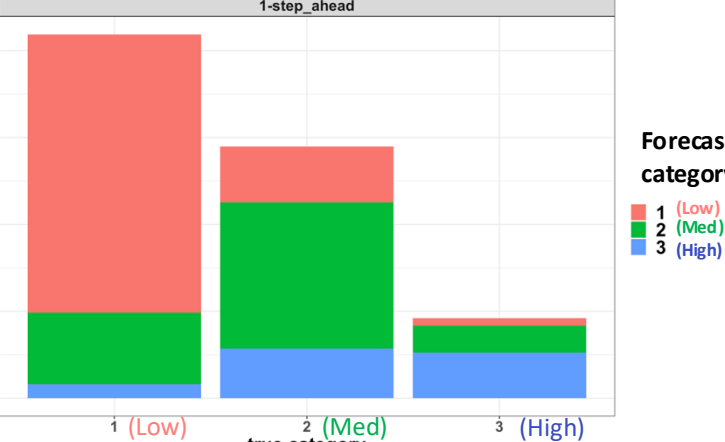


Fig : Overview of the combined district level forecasts. Comparison between 1-week ahead forecast categories and the true categories.

Temporal probabilistic forecasts and their evaluation using Brier score

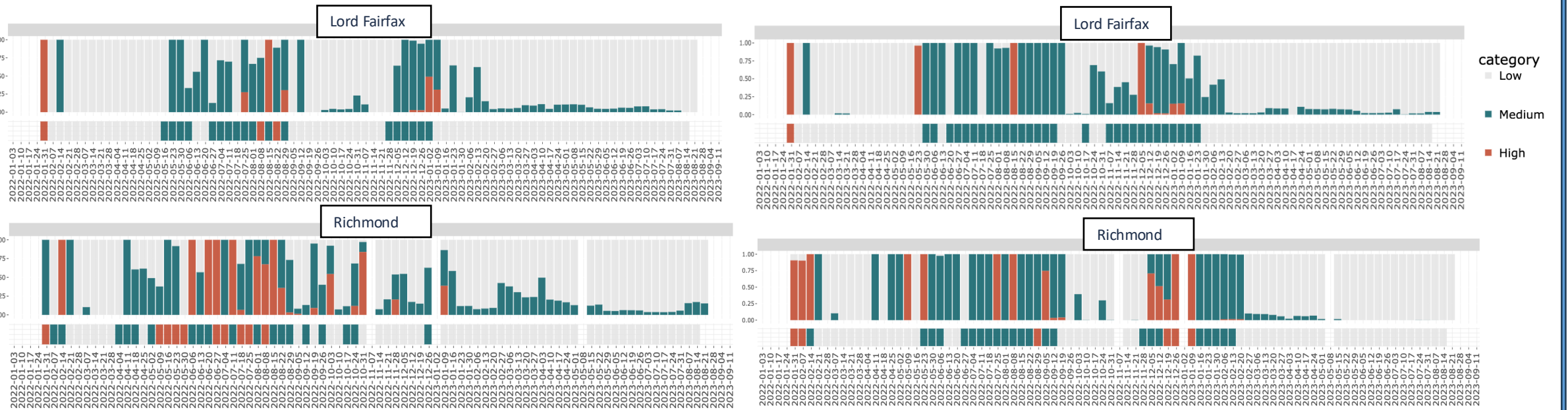


Fig : Probability forecasts and ground truth over time for two districts – Lord Fairfax and Richmond. Left side represents case forecasts and right side represents hospitalization forecasts.

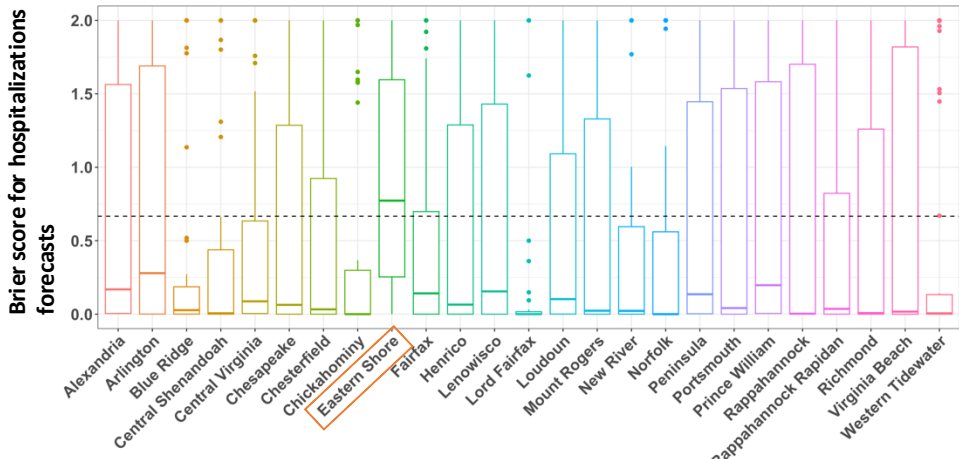
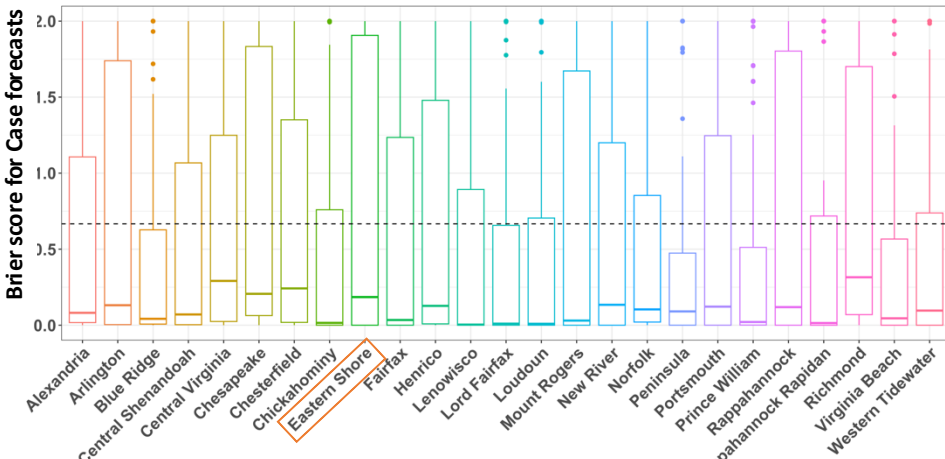


Fig: District level Brier scores evaluating probability forecasts for cases and hospitalization. The dotted line denoting $\frac{2}{3}$, corresponds the Brier Score of the Naïve model assigning equal probability for each category.

Observation & Challenges:

- Quantile categories lack robustness: highly sensitive to new values at any time
- Variable performance across districts due to varying data quality.

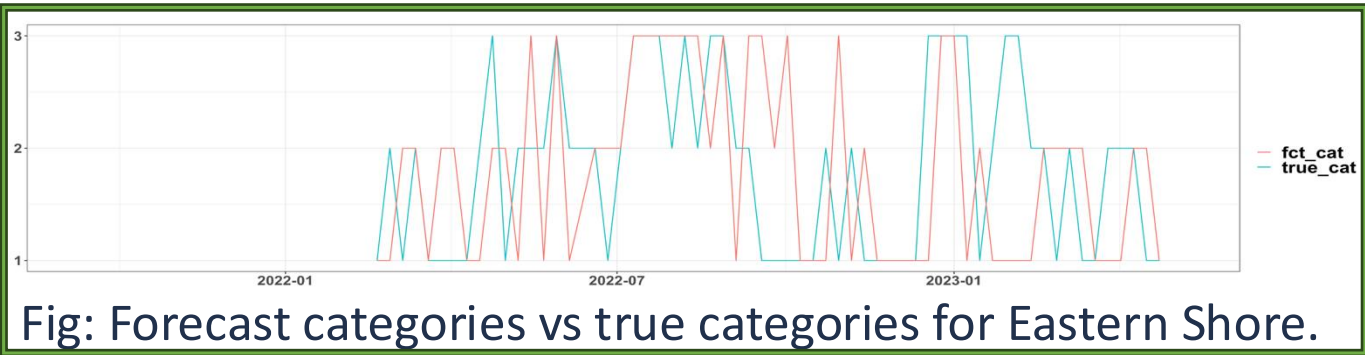


Fig: Forecast categories vs true categories for Eastern Shore.

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Evaluating utility of wastewater surveillance in multivariate forecasting for COVID-19

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VDH authors? – Rekha, Justin, Alex, ?
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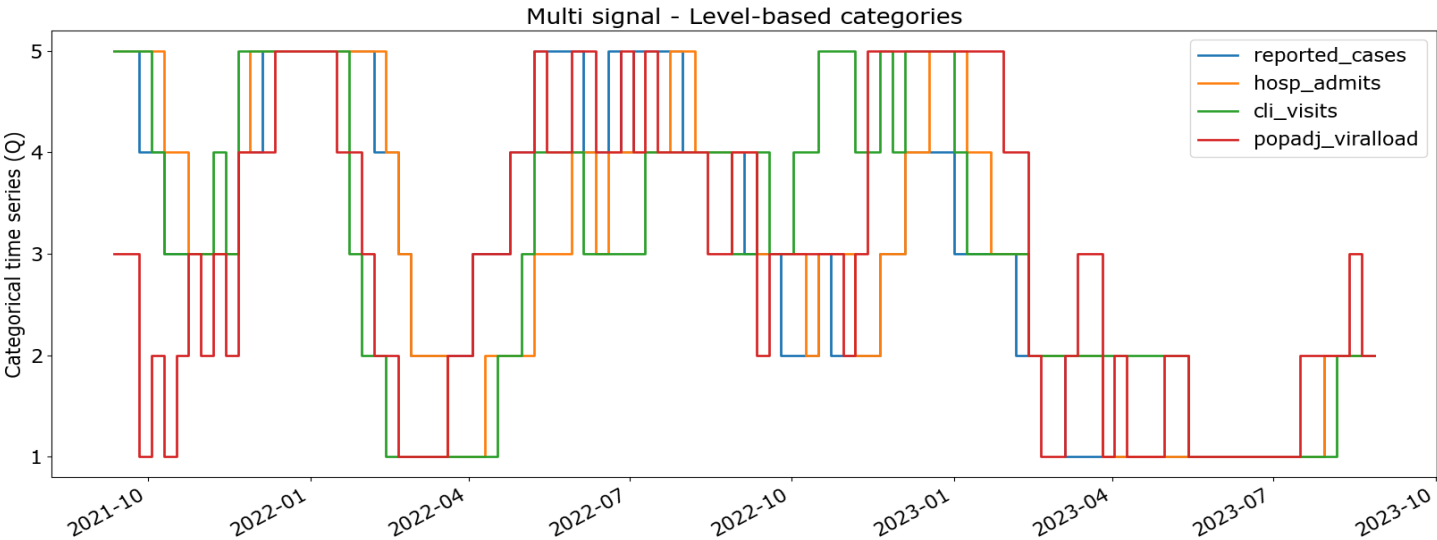
Motivation

Dataset Details

Spatiotemporal Alignment

Categorical Indicators:

Level-based categories: determining bin boundaries based on 5 quantiles



Multinomial regression formulation

$$\eta(Q_{cases}(t)) = \beta_0 + \beta_1 Q_{cases}(t-1)$$

4 logistic models for a categorical variable with 5 values:
for $k = 2,3,4,5$

$$\log\left(\frac{P(Q_{cases}(t) = k)}{P(Q_{cases}(t) = 1)}\right) = \beta_{1,0}^k + \beta_{1,1}^k I_{Q_{cases}(t-1)=1} + \beta_{k,2} I_{Q_{cases}(t-1)=2} + \beta_{k,3} I_{Q_{cases}(t-1)=3} + \beta_{k,4} I_{Q_{cases}(t-1)=4} + \beta_{k,5} I_{Q_{cases}(t-1)=5}$$

A multinomial model including all data sources: **C**ases
Hospitalizations, **CLI**, **W**astewater (CHIW)

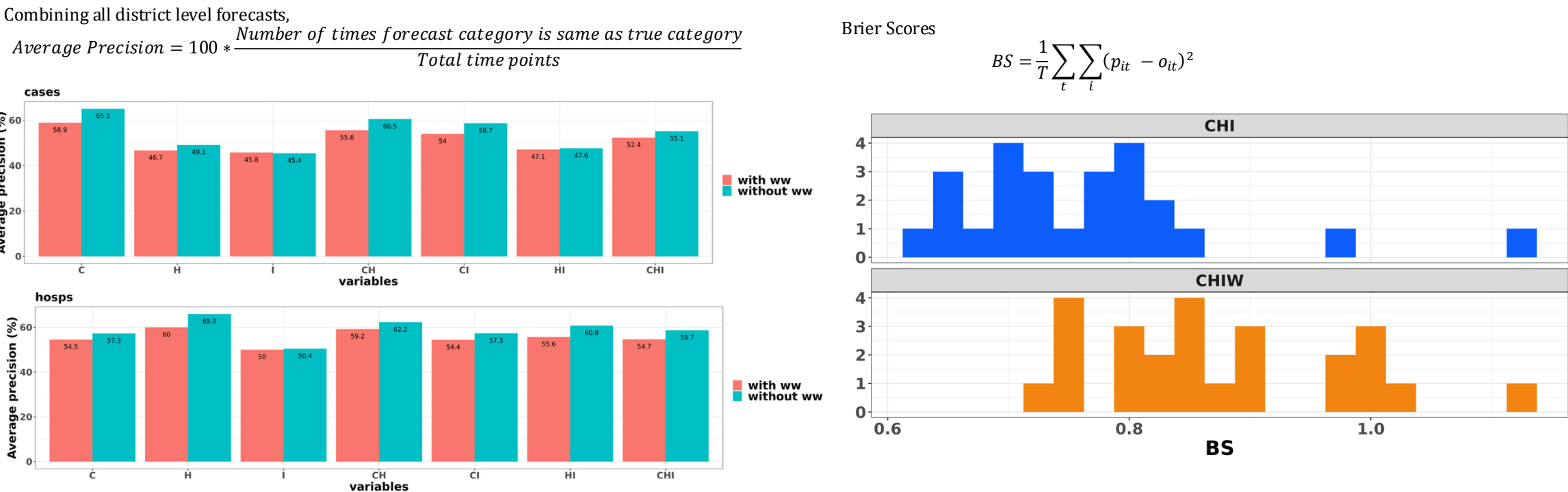
$$\eta(Q_{cases}(t)) = \beta_0 + \beta_1 Q_{cases}(t-1) + \beta_2 Q_{hosp}(t-1) + \beta_1 Q_{cli}(t-1) + \beta_1 Q_{ww}(t-1)$$

Evaluation Metrics

Figure for full experiment design

Results – Key Takeaways

Direct comparison with/without wastewater



Direct comparison – Cases vs Hospitalizations

Confusion matrix combining all districts, using model with all variables

$$\eta(Q_{cases}) = \beta_0 + \beta_1 Q_{cases} + \beta_1 Q_{hosp} + \beta_1 Q_{cli} + \beta_1 Q_{ww}$$

- Combining all districts, the average precision is higher for hospitalizations compared to cases
- Precision within the Higher and Highest category is 68.42 % for cases and 70.33% for hospitalizations.
-

Cases	Forecasted category					Total
	Lowest	Lower	Middle	Higher	Highest	
True Category						
Lowest	680 78.3 %	167 17.3 %	79 8.2 %	21 2.1 %	20 2.1 %	967 100 %
Lower	117 27 %	173 40 %	78 18 %	48 11.1 %	17 3.9 %	433 100 %
Middle	41 12.8 %	80 24.9 %	117 36.4 %	60 18.7 %	23 7.2 %	321 100 %
Higher	18 5.9 %	52 17 %	87 28.4 %	112 36.6 %	37 12.1 %	306 100 %
Middle	31 15.7 %	22 11.1 %	16 8.1 %	46 23.2 %	83 41.9 %	198 100 %
Total	887 39.9 %	494 22.2 %	377 16.9 %	287 12.9 %	180 8.1 %	2225 100 %

Average precision = 52.36 %

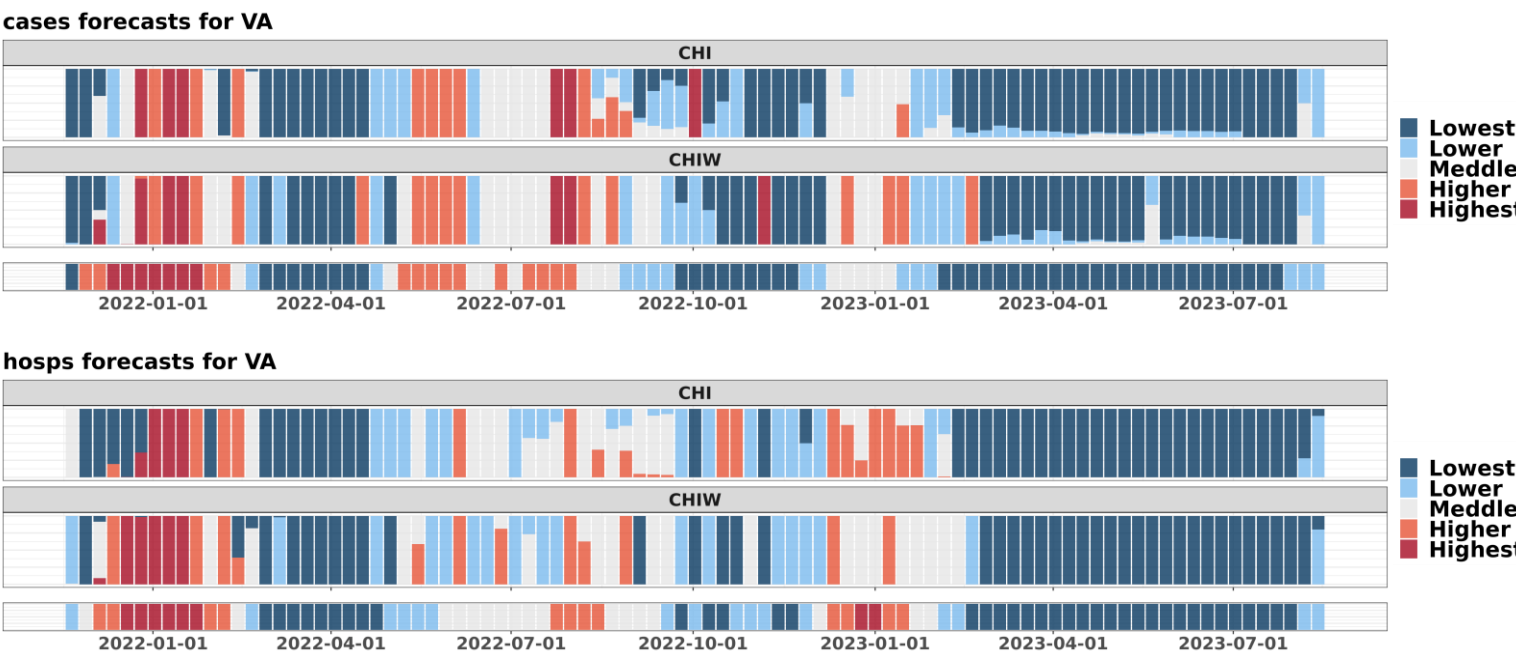
Hospitalizations	Forecasted category					Total
	Lowest	Lower	Middle	Higher	Highest	
True Category						
Lowest	676 78.5 %	126 14.6 %	31 3.6 %	17 2 %	11 1.3 %	861 100 %
Lower	120 25.3 %	195 41.1 %	109 23 %	40 8.4 %	10 2.1 %	474 100 %
Middle	33 9.1 %	110 30.3 %	134 36.9 %	73 20.1 %	13 3.6 %	363 100 %
Higher	33 11.1 %	40 13.5 %	90 30.3 %	99 33.3 %	35 11.8 %	297 100 %
Middle	32 13.9 %	14 6.1 %	17 7.4 %	55 23.9 %	112 48.7 %	230 100 %
Total	884 40.2 %	485 21.8 %	381 17.1 %	284 12.8 %	181 8.1 %	2225 100 %

Average precision = 54.65 %

Temporal probabilistic forecasts with ground truth

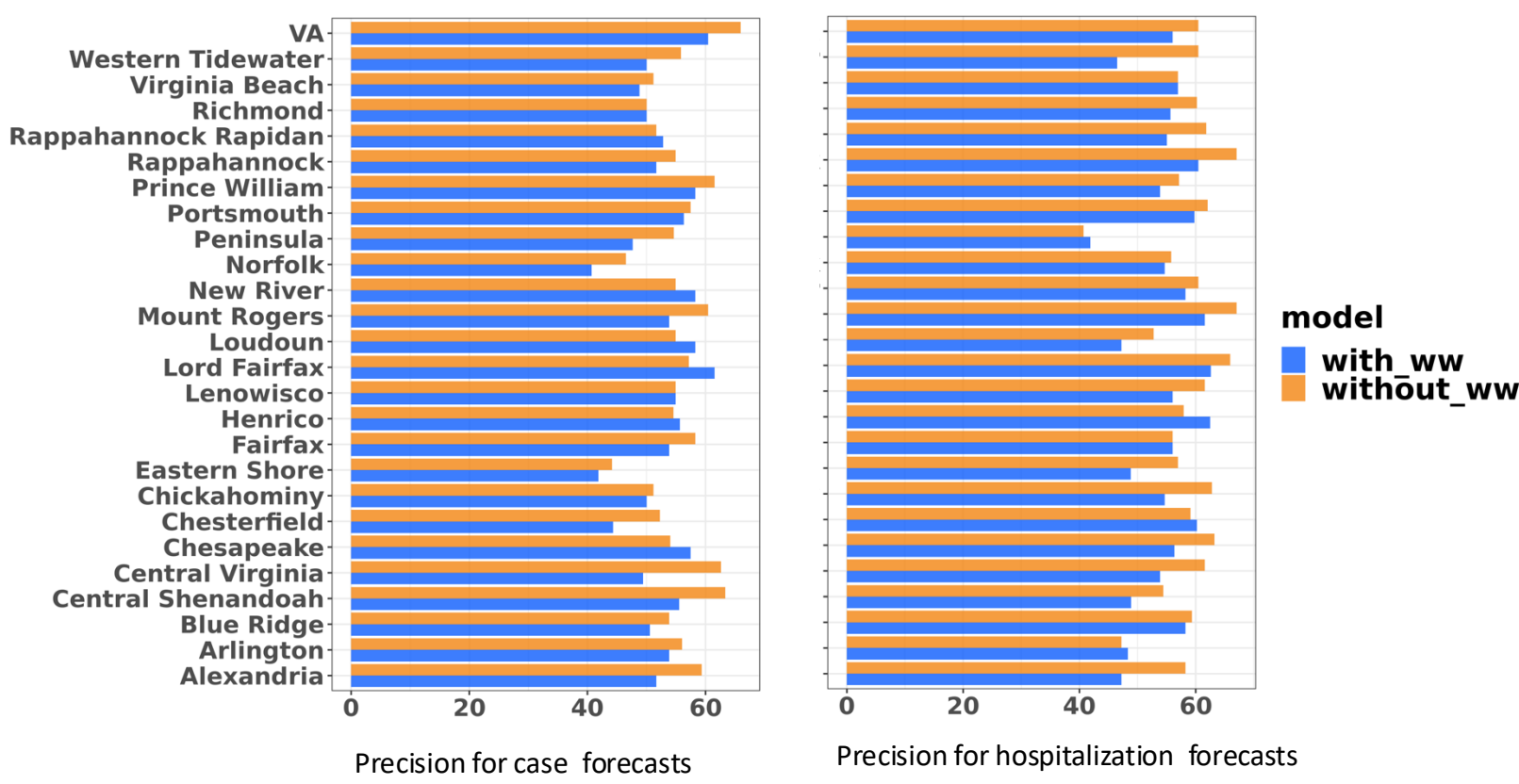
Using the full model with all variables:

$$\eta(Q_{cases}) = \beta_0 + \beta_1 Q_{cases} + \beta_1 Q_{hosp} + \beta_1 Q_{cli} + \beta_1 Q_{ww}$$



Impact of spatial resolution

State vs District level
forecast performance



Interactive tool for categorical indicators

References?

Acknowledgments

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